

> **"ONE HUNDRED MILLION SMALL FARMERS IN INDIA LOSE MONEY ON AGRICULTURE. IT'S CLEAR THEY NEED AFFORDABLE SOLUTIONS AGAINST CLIMATE RISKS"**
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> **KAUSHIK KAPPAGANTULU**

In **2015, when serial entrepreneur Kaushik Kappagantulu met Reshma,** a smallholder farmer in south India, he marvelled at the resilience of the spirit he was witnessing. Despite facing the worst of nature every time she turned to it for bounty, she refused to let her smile drop. That year, half her tomato crop had been wiped out by pests. The year before, unseasonal rains had destroyed her pepper. For six months every year, her land lay fallow because of how hot it was. "And Reshma is not alone," says Kappagantulu. "One hundred million small farmers in India on average lose money from agriculture. Meeting her made it clear that small farmers need affordable solutions against climate risks. That is what set us on the path to creating Kheyti."

Though he had experience uplifting India's marginalised through nifty and scalable ideas—his last gig had been getting high-skilled jobs for 50,000 school dropouts annually—he needed help. Alongside like-minded peers—Saumya (35), Ayush Sharma (39) and Satya Raghu Mokkapati (39)—he began talking to farmers across the country to figure out what ailed them. Drawing from his engineering and business training at IIT Kharagpur and Columbia Business School, he came up with an idea: a modular greenhouse, a fenced oasis on a small part of the farmer's land, tailored to the region's unique climate and able to thwart adverse conditions by harnessing available resources with methodical efficiency.

But there were challenges. Though the greenhouse was theoretically sound—the enclosure would be around 1/16th of an acre, with netting to cut pest attacks, shade to reduce heat and a facility for drip irrigation to conserve water—it was too expensive for farmers. Initial support would be philanthropic, but they would eventually need to be able to buy the greenhouse, learn the technologies involved and also get daily support. The biggest obstacle? They had to be convinced that such an unconventional solution was worth it.

Over the next few years, however, after taking feedback from thousands of farmers across diverse climatic regions like Telangana, Andhra Pradesh, Madhya Pradesh, Odisha, Jharkhand, Bihar, Rajasthan and Uttar Pradesh—including arid and tribal parts—Kheyti began to flourish. The team redesigned the greenhouse so that it could be built with local materials, cutting costs by 80 per cent. Heat impact was reduced by nearly 13 degrees Celsius. Water was conserved by up to 98 per cent. Early success stories and trusted local teams with meticulous tracking routines built faith. A greenhouse crop cycle yielded up to seven times more than what you got with the same crop in traditional open-field farming. Farmer incomes have gone up by two to four times, depending on crop choice, consistency of greenhouse usage and access to markets—and because farmers can also grow off-season, they capture better prices. Three states—Andhra, Telangana and Uttar Pradesh—have even incorporated the model in government subsidy programmes. While 7,000 smallholder farmers around the country are working with Kheyti today, the goal is to reach a million by 2030.

"What makes Kheyti's model deliver is not just the technology, but the trust built with farmers over time," says Kappagantulu. "The solution was not imposed from outside. It was co-created with farmers through years of listening and testing... Through Kheyti, smallholder farmers are not only adapting to climate change. They are leading the way in solving it." ■

FACT SHEET

Name of company
Kheyti

What it does **Builds greenhouses for farmers to protect their produce from climate shocks**

Founded in **2015**

Annual turnover
Rs 8.3 crore